

API Usage & Integration Guide

LOGISTICS OBJECT DETECTION & REASONING REST API

1. Base URLs & Swagger Documentation

- Live AWS Production Base URL: <http://13.233.255.22>
- Interactive Swagger UI Console: <http://13.233.255.22/docs>
- Local Server Fallback: <http://localhost:7860>

2. Endpoint Summary Table

Endpoint	HTTP Method	Content-Type	Description
/detect	POST	multipart/form-data	Returns bounding boxes, class names, and confidence scores with 4-stage post-processing guardrails.
/ask	POST	multipart/form-data	Natural language question answering with 3-tier confidence floor guardrail.
/health	GET	application/json	Liveness probe returning model load status.
/classes	GET	application/json	Returns the 5 supported industrial object categories.

3. Detection Endpoint (POST /detect)

cURL Example:
`curl -X POST "http://13.233.255.22/detect" \ -H "Accept: application/json" \ -F "file=@warehouse_photo.jpg"`

Sample Response Payload (200 OK):
`{ "objects": [ { "class": "freight container", "confidence": 0.5321, "bbox": { "x1": 76.53, "y1": 23.41, "x2": 589.8, "y2": 597.75 } } ], "num_detections": 1, "image_size": [640, 640] }`

4. Reasoning & Question Endpoint (POST /ask)

cURL Example:
`curl -X POST "http://13.233.255.22/ask" \ -H "Accept: application/json" \ -F "file=@warehouse_photo.jpg" \ -F "question=How many freight containers are visible?"`

Sample Response Payload (High Confidence Answer):
`{ "answer": "There is 1 freight container visible.", "used_detector": true, "confidence": "high", "detections": [ { "class": "freight container", "confidence": 0.5321, "bbox": { "x1": 76.53, "y1": 23.41, "x2": 589.8, "y2": 597.75 } } ], "intent": "COUNT" }`

5. Python Integration Example

```
import requests url = 'http://13.233.255.22/ask' files = {'file': open('warehouse.jpg', 'rb')} data = {'question': 'Is there a forklift near the container?'} response = requests.post(url, files=files, data=data) print(response.json())
```